

On the relationship between privative and equipollent feature specifications*

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1 Introduction

Since the time of Trubetskoy (1939) and Jakobson (1941), phonologists have pondered whether feature specifications for segments and natural classes ought to be defined using *privative* or *equipollent* (i.e., binary-valued), or other possibilities. This study formally compares equipollent and privative feature specifications by determining the conditions under which they are or are not equivalent. A bijection is proven between *consistent* and *complete* (in a sense to be defined) equipollent and privative **segments**, grounding the intuition that under- and contradictory specifications have no equivalent in privative systems. Then, it is shown that there is no isomorphism between equipollent and privative **natural classes**, grounding the intuition that the former defines natural classes inexpressible in the latter.

Some prior formal work has considered these questions. Nelson (2022) shows that privative natural class specifications define the same natural classes as equipollent specifications, but only if these natural class specifications can refer to absence, something contrary to common assumption (e.g., Lees 1961, Lightner 1971, Archangeli 1988, Reiss 2003, Bale et al. 2020). Danis (2025) proves that two feature theories define the same segment inventories but different natural classes. This study generalizes these results to arbitrary sets of feature predicates and the common-place interpretation of natural classes using proofs of bijection and/or isomorphism.

I first provide some basic definitions. Then in section 2, I give a bijection between equipollent and privative features, proving the conditions under which it is valid. In section 3, I define the extension operation over natural classes, proving that there exists no isomorphism between equipollent and privative features which preserves it. Linguistic implications of these findings are discussed in section 4.

Definition 1.1. A function $\phi : A \rightarrow B$ is *injective* iff $\forall a_1, a_2 \in A, \phi(a_1) = \phi(a_2)$ implies $a_1 = a_2$.

Definition 1.2. A function $\phi : A \rightarrow B$ is *surjective* iff $\forall b \in B, \exists a \in A$ such that $\phi(a) = b$.

*Thanks to Scott Nelson for helpful comments.

Definition 1.3. A function is *bijective* iff it is injective and surjective.

In other words, a function from A to B is bijective iff each element of A is paired with exactly one element of B and vice versa.

Definition 1.4. A function $\phi : A \rightarrow B$ is *structure-preserving* w.r.t. an operator $\bullet$ if $\forall a_1, a_2 \in A$, $\phi(a_1 \bullet a_2) = \phi(a_1) \bullet \phi(a_2)$.

Definition 1.5. Finally, A, B are *isomorphic* w.r.t. $\bullet$ iff there exists a structure-preserving bijection $\phi : A \rightarrow B$.

2 Segments

We first consider the segment inventories defined by privative versus equipollent feature systems. Let F be a finite, non-empty set of binary-valued feature predicates, and let $\mathcal{P}$ be the power set function. Let $\mathbb{B} = \mathcal{P}(\{+, -\} \times F)$ be the universe of equipollent feature specifications. Then, an equipollent segment is an element of $\mathbb{B}$.

Definition 2.1. An equipollent segment S is *consistent* if $\forall f \in F$, $+f \in S$ implies $-f \notin S$ and $-f \in S$ implies $+f \notin S$; that is, contradictory specification is not permitted.

Definition 2.2. An equipollent segment S is *complete* if $\forall f \in F$, $+f \in S \vee -f \in S$; that is, underspecification is not permitted.

Let $\mathbb{B}_{cc}$ be the consistent and complete subset of $\mathbb{B}$; i.e., such that for every segment S in $\mathbb{B}_{cc}$, $\forall f \in F$, $+f \in S \underline{\vee} -f \in S$ where $\underline{\vee}$ denotes logical exclusive or. Let $\mathbb{P} = \mathcal{P}(F)$ be the universe of privative feature specifications. Then a privative segment is a subset of F . Example segments so defined are shown below.

(1) Segments as a sets of features:

Equipollent:

Privative:

$$\left\{ \begin{array}{l} +\text{SYLLABIC} \\ +\text{BACK} \\ +\text{ROUND} \\ +\text{HIGH} \\ -\text{Low} \\ +\text{ATR} \end{array} \right\} = \left\{ \begin{array}{l} \text{SYLLABIC} \\ \text{BACK} \\ \text{ROUND} \\ \text{HIGH} \\ \text{ATR} \end{array} \right\} = /u/$$

We now make this bijection precise:

Theorem 2.1. Let $\phi : \mathbb{B}_{cc} \rightarrow \mathbb{P}$ where $\phi(B) = \{f \in F \mid +f \in B\}$. ϕ is a bijection between $\mathbb{B}_{cc}$ and $\mathbb{P}$.

Proof. (Injectivity.) Suppose $B_1, B_2 \in \mathbb{B}_{cc}$ and $\phi(B_1) = \phi(B_2)$. By the definition of ϕ , $\forall f \in F$, $+f \in B_1 \iff +f \in B_2$, and by consistency and completeness, $\forall f \in F$, $-f \in B_1 \iff -f \in B_2$. Thus $B_1 = B_2$. (Surjectivity.) Suppose $P \in \mathbb{P}$, and let $B = \{+f \mid f \in P\} \cup \{-f \mid f \in F \setminus P\}$. By construction B is consistent and complete; i.e., every feature is paired with exactly one coefficient. Thus $\phi(B) = P$. By injectivity and surjectivity, ϕ is a bijection. $\square$

The following remarks justify the definition of ϕ with the consistent and complete domain $\mathbb{B}_{cc}$.

Remark 2.1. Consistency is a necessary condition on the domain of ϕ . Suppose that $F = \{\text{CONTINUANT}\}$, and let $B = \{+\text{CONTINUANT}\}$ and $B' = \{+\text{CONTINUANT}, -\text{CONTINUANT}\}$; the former is consistent but the latter is not. Then, if we extend ϕ 's domain from $\mathbb{B}_{cc}$ to $\mathbb{B}$, $\phi(B) = \phi(B') = \{\text{CONTINUANT}\}$. Because $B \neq B'$, ϕ is not injective and thus not bijective.

Remark 2.2. Completeness is also a necessary condition on the domain of ϕ . Let $n = |F|$. Each feature $f \in F$ may appear in a consistent equipollent specification in three ways: as $+f$, as $-f$, or not at all. By consistency, these are mutually exclusive, so there are 3^n consistent specifications. Privative specifications are subsets of F , of which there are 2^n . Since $3^n > 2^n$ for $n \geq 1$, no injection (or bijection) between consistent equipollent and privative specifications can exist without the additional completeness condition.

3 Natural classes

Following common practice (e.g., Bale et al. 2020), let us assume that natural classes are also defined via feature specifications, but their extension, denoted by $[\bullet]$, is the superset relation, so that the extension of a natural class is essentially a set of segments (i.e., a set of sets of feature specifications).

Definition 3.1. For equipollent specifications $[B_0] = \{B \in \mathbb{B} \mid B \supseteq B_0\}$.

Definition 3.2. For privative specifications $[P_0] = \{P \in \mathbb{P} \mid P \supseteq P_0\}$.

The following shows an example of extension as applied to equipollent and privative specifications.

$$(2) \quad \begin{array}{ll} \text{a. Equipollent:} & [+HIGH] = \{x \mid x \supseteq \{+HIGH\}\} \\ \text{b. Privative:} & [HIGH] = \{x \mid x \supseteq \{HIGH\}\} \end{array} = \{i, y, i, u, u, \dots\}$$

Let $\mathcal{C}_B$ and $\mathcal{C}_P$ denote the set of all equipollent and privative natural classes, respectively.

Remark 3.1. Equipollent natural classes do not explicitly require consistency, but inconsistently specified natural classes denote only the empty set under the above definitions, so it is assumed here, without loss of generality, that equipollent natural classes are consistent.

Remark 3.2. Neither equipollent nor privative natural classes require completeness, because partial specification is necessary to allow natural classes to extend over more than a single segment.

Theorem 3.1. There is no isomorphism between $\mathcal{C}_B$ and $\mathcal{C}_P$ preserving $[\bullet]$.

Proof. Let $n = |F|$. Privative natural classes are in a bijection with subsets of F , so $|\mathcal{C}_P| = 2^n$. Equipollent natural classes correspond to consistent subsets of $\{+, -\} \times F$, so $|\mathcal{C}_B| = 3^n$. Since $3^n > 2^n$ for $n \geq 1$, no injection (or bijection, or isomorphism) exists. $\square$

4 Implications

Consistency is not a completely uncontroversial assumption for those who assume equipollent features. In Basque (Hualde 1991: ch. 5), affricates pattern with stops with respect to a process which deletes the first of two adjacent stops: $/...t\hat{s}-t.../ \rightsquigarrow [...zt...]$ as in *hiztegi* ‘dictionary’. On the other hand, also they pattern with sibilant fricatives with respect to a separate process which simplifies an affricate before a sibilant fricative: $/...t\hat{s}-t\hat{s}.../ \rightsquigarrow [...st...]$ as in *irakasten* ‘teach’ (IMPF.) from Hualde thus suggests that Basque affricates are both +CONTINUANT and –CONTINUANT, so that the former rule targets –CONTINUANT stops and the latter targets +CONTINUANT fricatives. Remark 2.1 grounds the intuition that an analysis depending on inconsistent specification cannot be expressed under privative specification barring segment-internal ordered featural structure.

Many theories of phonology assume equipollent feature specification but permit underspecification, in violation of completeness. Perhaps one of the most compelling approaches to underspecification is *archiphonemic underspecification*. Inkelas and colleagues (e.g., Inkelas and Cho 1993, Buckley 1994, Inkelas 1995, Inkelas et al. 1997) use this to generate a pattern they call *inalterability as prespecification* (IAP). In this pattern, segments underspecified for some feature f are *mutable* with respect to feature-filling rules supplying $+f$ or $-f$, whereas segments prespecified as $+f$ or $-f$ are *inalterable* with respect to those same rules. Kiparsky (1993) and Rasin (in press) generalize this pattern to derive non-derived environment blocking, and Gorman and Reiss (in press) further generalize this pattern from target segments to trigger segments. IAP is commonly used to generate “three-way” segmental contrasts like found in Turkish (Inkelas and Orgun 1995):

		NOM.SG.	ACC.SG.	
(3)	a.	/sanat-/:	sanat	sanat-i ‘art’
	b.	/etyd-/:	etyd	etyd-y ‘etude’
	c.	/metoD-/:	metot	metod-u ‘method’

Inkelas and Orgun argue that the final consonant of ‘method’, written /D/, is underspecified for VOICE, but otherwise identical to /t, d/. This underspecification makes it mutable with respect to a feature-filling intervocalic voicing process. Note that it is not possible to treat the root morpheme of ‘method’ as exceptionally undergoing intervocalic voicing, since the root’s other oral stop is consistently realized as intervocalic [t]. Similar patterns are found in Île de Groix Breton (Krämer 2000) and Bakairi (Wetzels and Mascaró 2001). Remark 2.2 grounds the intuition that IAP is fundamentally incompatible with privative specification, as conjectured by Inkelas (1995).

Suppose that $F = \{\text{VOICE}\}$. Then $\mathcal{E}_B$ contains three natural classes: voiced segments [+VOICE], voiceless segments [–VOICE], and the “universal” class, written []. In contrast, $\mathcal{E}_P$ contains just two: voiced segments [VOICE] and the universal []. In such a system, there is no counterpart to the equipollent [–VOICE]. Theorem 3.1 is simply a generalization of this result. Yet some phonologists (e.g., Iverson and Salmons 1995, Lombardi 1995a,b) argue that it is desirable to treat VOICE as privative. How then does they derive the common pattern of voicelessness assimilation? One common technique is to permit processes to be triggered by natural classes which refer, directly or indirectly, to the absence of specific features, often in an autosegmental framework. For example, Kiparsky (1985) proposes novel notation for referring to an absence of particular association lines. Similarly, Lombardi (1995b) assumes that privative VOICE specifications are part of a separate laryngeal node and derives voicelessness assimilation by allowing laryngeal nodes not specified

for VOICE to spread, triggering delinking on the target. Future work should consider the formal relationship between such strategies and the formalism developed here.

5 Conclusions

Under general conditions, consistent and complete equipollent and privative feature specifications define equivalent sets of segments, but equipollent feature systems define a larger set of natural classes than privative systems. These results are independent of the choice feature predicates and any assumptions about featural markedness. These largely intuitive results have implications for our understanding of affricates, the “archiphonemic” use of underspecification to derive three-way contrasts, and the assimilation of laryngeal features, suggesting that these phenomena will be key to the choice between equipollent and privative feature specification. While the above discussion focuses on phonology, many of the same issues arise in realizational theories of morphology (e.g., Zwicky 1985, Halle and Marantz 1993) which have rules of *exponence* and/or *referral* defined by privative and/or equipollent natural class targets.

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